

Abhishek Milind Patwardhan

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EXPERIENCE

Software Development Engineer I | Syspro Technologies | Plano, TX, USA

Aug 2025 – May 2026

- Implemented a Claude API-powered LLM assistant to analyze daily build reports, cutting analysis time from 58 to 14 minutes and saving the team about 44 minutes every morning.
- Worked on Payment & Test microservices (Spring Boot, Spring MVC, JPA, Hibernate, REST) handling 1,000+ concurrent requests, as part of an Agile/Scrum team.
- Contributed to migrating 8 REST endpoints to GraphQL to support mobile clients without duplicating APIs, reducing query response time by 25% by removing redundant legacy fields.
- Set up Jenkins and GitHub Actions CI/CD pipelines for nightly regression testing on Linux build agents, saving the team roughly 40+ hours a month of manual verification work; also wrote SQL queries to help debug performance regressions.

Graduate Research Assistant | University of Georgia | Athens, GA, USA

Aug 2024 – Jun 2025

- Worked on an adaptive learning engine (NCD model) that recommended videos based on each student's weak topics; 80% of students in the adaptive cohort scored higher on their second assessment, compared to a self-paced control group.
- Fine-tuned a Phi-3 LLM on 25 instructional videos as part of my research, improving accuracy on context-restricted questions from 65% to 85%, tested against more than 300 prompts including edge cases generated using Claude.
- Helped deploy and maintain a containerized Django-PostgreSQL backend on AWS EC2 (Linux) using Docker and Kubernetes for service orchestration, supporting nearly 2,000 API requests daily for 100+ students across 20 topics with role-based authentication.
- Contributed to a role-gated EdTech platform with adaptive and self-paced cohorts, built across 4 iterative sprints.

PROJECTS

StopByShow: Movie Ticket Booking Platform | Spring Boot, MySQL, Docker, RabbitMQ, JWT, OAuth2.0 | [GITHUB](#)

- Developed 11 microservice modules using an event-driven architecture with RabbitMQ, supporting nearly 10,000 concurrent users.
- Used Spring Boot's dependency injection (IoC) and Spring Data JPA to keep the service layers clean and maintainable; secured 10+ endpoints using JWT, RBAC, and OAuth2.0.
- Wrote unit and integration tests with JUnit and Mockito, achieving 85% coverage across all service layers.

Restaurant Review Platform | Spring Boot, Spring Security, Keycloak, Elasticsearch, TypeScript | [GITHUB](#)

- Secured 6 endpoints with Spring Security and Keycloak using role-based access control for user and admin roles.
- Integrated Elasticsearch for full-text search by pincode, locality, and cuisine, cutting search time by 65% over direct DB queries.

Consistent Hashing Naming Service | Java, TCP Sockets, Distributed Systems | [GITHUB](#)

- Built a distributed naming service across 10+ nodes with TCP sockets, handling node entry, exit, & key transfer for service discovery.
- Tested and confirmed near-optimal 1/N key redistribution per topology change, across 500+ key-value pairs.

Persistent Async Multicast System | Java, Distributed Systems, Multithreading | [GITHUB](#)

- Engineered a multithreaded coordinator-participant system to ensure zero message loss for nodes that reconnect after disconnecting.
- Verified delivery integrity across buffer thresholds of 5 seconds, 1 minute, and 5 minutes.

TECHNICAL SKILLS

Languages: Java (Core, Multithreading, Concurrency, OOP, JVM), Python, JavaScript, TypeScript

Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST API Design, GraphQL, Django

AI/ML: Claude API, Phi-3 Fine-Tuning, Adaptive Learning Models

Databases & System Design: PostgreSQL, MySQL, MongoDB, Redis, Distributed Systems, Event-driven Architecture, Message Queues, Load Balancing, Fault Tolerance, Service Discovery, Design Patterns

Tools & Practices: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, RabbitMQ, Linux, Bash, Agile, SDLC, JUnit, Mockito

EDUCATION

M.S. Computer Science, University of Georgia, USA (GPA - 3.95/4.0)

Aug 2023 – Aug 2025

Relevant Coursework : Distributed Systems, Software Engineering, Algorithms, ML Optimization

B.E. Computer Engineering, University of Mumbai (GPA - 8.97/10.0)

Aug 2019 – May 2023

Relevant Coursework : Data Structures, DBMS, Artificial Intelligence, Object Oriented Programming

CERTIFICATIONS

- AWS Certified Cloud Practitioner, Amazon Web Services